

COURSE TITLE : COMMUNICATION ENGINEERING
COURSE CODE : 6241
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Radiation and wave propagation	15
2	Modulation Techniques	15
3	Radio Transmitters	15
4	Radio Receivers	15
TOTAL		60

Course General Outcome:

MODULE	G.O.	ON COMPLETION OF THIS COURSE THE STUDENT WILL BE ABLE:
1	1	To understand the Electro Magnetic Radiation and wave propagation
	2	To understand Antennas
2	3	To comprehend Modulation Techniques in Communication
	4	To understand pulse modulation
3	5	To understand the working of Radio Transmitters
	6	To understand noise in Communication Systems
4	7	To understand demodulation and Radio receivers

On completion of this course the student will be able:

MODULE I RADIATION AND WAVE PROPAGATION

1.1.0 To understand Electro Magnetic Radiation and Wave Propagation

- 1.1.1 To define electric and magnetic fields.
- 1.1.2 To define EM wave
- 1.1.3 To explain types of EM wave
- 1.1.4 To list the types of wave propagation
- 1.1.5 To explain ground wave propagation
- 1.1.6 To explain the space wave propagation
- 1.1.7 To explain layers of Ionosphere, its significance in communication
- 1.1.8 To define- skip distance, max usable frequency (MUF), critical frequency, virtual height

1.2.0 To understand Antennas

- 1.2.1 To explain the physical concept of radiation of electromagnetic energy.
- 1.2.2 To describe radiation patterns
- 1.2.3 To explain the different types of antenna such as Half wave dipole, folded dipole

- 1.2.4 To explain the working of parabolic antenna
- 1.2.5 To define smart antenna and its applications
- 1.2.6 To define MANET

MODULE II MODULATION TECHNIQUES

2.1.0 To comprehend Modulation Techniques in Communication

- 2.1.1 To state the need for modulation
- 2.1.2 To define Amplitude modulation, frequency modulation and phase modulation
- 2.1.3 To deduce expression for amplitude modulated wave
- 2.1.4 To define modulation index of AM
- 2.1.5 To describe frequency spectrum of AM
- 2.1.6 To explain the AM modulator circuits (collector modulation)
- 2.1.7 To explain balanced modulator
- 2.1.8 To define DSBSC, SSBSC and VSB
- 2.1.9 To draw the frequency spectrum of DSBSC, SSBSC and VSB
- 2.1.10 To define modulation index of FM
- 2.1.11 To describe Frequency spectrum, band width, related with FM

2.2.0 To understand pulse modulation

- 2.2.1 To state the sampling theorem and its significance
- 2.2.2 To list different pulse modulation techniques
- 2.2.3 To discuss Pulse Code Modulation
- 2.2.4 To explain quantization
- 2.2.5 To explain companding
- 2.2.6 To list different digital carrier modulation schemes

MODULE III RADIO TRANSMITTERS

3.1.0 To understand the working of Radio Transmitter

- 3.1.1 To explain the block diagram of AM transmitter
- 3.1.2 To explain the block diagram of FM transmitter (direct FM and indirect FM)
- 3.1.3 To define pre-emphasis and de-emphasis
- 3.1.4 To describe AFC

3.2.0 To understand Noise in Communication system

- 3.2.1 To list different types of noises

MODULE IV RADIO RECEIVERS

4.1.0 To understand the need for demodulation and radio receivers

- 4.1.1 To define demodulation
- 4.1.2 To explain super heterodyne receiver
- 4.1.3 To explain AM demodulation circuits using diode detector
- 4.1.4 To explain simple and delayed AGC
- 4.1.5 To define selectivity, sensitivity, fidelity and noise figure of Radio receivers
- 4.1.6 To explain the operation of FM radio receiver
- 4.1.7 To state the need of limiter in FM receiver
- 4.1.8 To compare AM & FM receivers

CONTENT DETAILS

MODULE - I ELECTROMAGNETIC RADIATION AND WAVE PROPAGATION

Electric and magnetic fields – electromagnetic wave - polarization types of EM wave-Types of wave propagation – ground wave propagation – Space wave propagation – Ionosphere and its layers – layer characteristics – skip distance- M.U.F – critical frequency – virtual height.

Antennas

Physical concepts of radiation of electromagnetic wave - radiation pattern - Half wave dipole, folded dipole and parabolic antenna-smart antenna -applications-MANET

MODULE - II MODULATION TECHNIQUES

Need for modulation: - basic principle of different types of modulation AM, FM, & PM. Definition of AM – expression for AM voltage wave form- modulation index significance of side bands-frequency spectrum - the basic AM modulator circuits-balanced modulator – definition of DSBSC,SSBSC and VSB- frequency spectrum of DSBSC,SSBSC and VSB- modulation index of FM signal-frequency spectrum and band width of FM- sampling theorem - different pulse modulation techniques-PAM,PWM,PPM -PCM- quantization-companding – digital carrier modulation schemes-ASK,PSK,FSK

MODULE – III RADIO TRANSMITTERS

Block diagram of AM transmitter- function of each block – Block diagram of a FM transmitter- Direct and indirect FM generation-pre-emphasis and de-emphasis-AFC- different types of noise in communication systems

MODULE – IV DEMODULATION AND RADIO RECEIVERS

Need for demodulation--Principles of superheterodyne receiver- Block diagram of superheterodyne receiver–AM demodulator circuits-diode detector-simple and delayed AGC - choice of IF in super heterodyne receiver – Characteristics of radio receiver- selectivity,sensitivity,fidelity and noise figure.

Block diagram of FM receiver – need for limiter in FM receiver-- Comparison of FM and AM receiver

TEXT BOOKS

1. Electronic Communication Systems - Frank R Dungan - 3rd Edition – Thomson.
2. Electronic Communication Systems – George Kennedy – TMH.

REFERENCE

1. Roddy and Coolen – Electronic communications - PHI.
2. Blake - Electronic Communication Systems - Thomson and Delmar.
3. Wayne Thomasi -Electronic Communication Systems. Pearson Education
4. Pradeep Kumar Ghosh -Principles of Electronic Communications CRC